

Antoine Boucher

SOFTWARE ENGINEER — PLATFORM AND COMPUTER GRAPHICS

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Summary

Cloud-native software engineer focused on platform engineering and computer graphics. Ship Kubernetes, GCP, Terraform, GitLab CI/CD, and observability for Java/Kotlin/Spring Boot and Python microservices; harden releases with JUnit, Playwright, and DevSecOps practices. Master's at ÉTS with real-time rendering, physics simulation, and tactile sensing in C++, OpenGL/GLSL, and Blender.

Skills

Graphics / 3D	Computer Graphics, OpenGL, GLSL, Real-time rendering, Physics simulation, C++, Blender
Backend	Java, Kotlin, Spring Boot, Python, REST APIs, GraphQL, Microservices
Frontend	Angular, React, Vue
DevSecOps / Cloud	Docker, Kubernetes, Terraform, IaC, GitLab CI/CD, GitHub Actions, Observability, GCP, Azure, AWS
Data / Storage	PostgreSQL, MySQL, MongoDB, Redis, Kafka, Data Science, Business Research
E2E / Testing	E2E testing (Playwright, Selenium), JUnit, ELK, Grafana, Prometheus, Sentry, Serilog
Other	Linux, Bash, Ansible, OWASP, Secure coding, SCRUM, SAFe, MCP
Credentials	AWS Certified Cloud Practitioner; QcES; AWS Cloud Quest

Experience

IMC2 – Institut multidisciplinaire en cybersécurité et cyberrésilience

Montréal, QC (Remote)

Research Assistant — Secure cloud platforms (Permanent Part-time)

Mar 2025 – Feb 2026

- Owned 5 GitLab CI/CD pipelines (split a monolith repo into per-service micro-repos) and runners across two GKE clusters on GCP, with Terraform-managed networking and observability, tightening repeatable cloud releases
- Containerized workloads in Docker; configured and operated the research GKE clusters on GCP (workload networking); rolled out services with Terraform, making infrastructure changes safer and faster for the team
- Hardened 5 Java microservices with JUnit and Playwright; promoted Android builds through GitLab into the Terraform-managed GKE environment, raising pre-release coverage and syncing mobile with backend delivery
- Shipped scripted cybersecurity risk analysis across 1000+ risks and 181+ threat categories with weighted scoring, star-based prioritization, and an Angular triage UI so analysts triage faster with consistent context
- Technical Skills:** Java, Kotlin, Android, Docker, Kubernetes, Terraform, GitLab CI/CD, GCP, JUnit, Playwright, Angular

Polytechnique Montréal

Montréal, QC (On-site)

Teaching Assistant — DevSecOps (LOG 8100) (Contract Part-time)

Sep 2024 – Jan 2026

- Built and maintained the GitLab workspace (repos, CI/CD templates, runners, workflows) for approximately 20 concurrent teams, reducing admin friction during the term
- Coached Docker and Kubernetes build, deploy, and debug under LOG 8100 goals; anchored security with OWASP-driven examples, connecting lecture theory to exploitable scenarios
- Refreshed DevSecOps-aligned lab artifacts, graded practicals, and coached resubmissions, improving submission quality and turnaround
- Technical Skills:** GitLab, CI/CD, DevSecOps, Docker, Kubernetes, OWASP, Secure Coding, Linux

École de technologie supérieure (ÉTS)

Montréal, QC (Hybrid)

Teaching Assistant (Contract Part-time)

Feb 2024 – Apr 2026

- Delivered labs, integrator-project support, and office hours on mobile/UX, integrator-project, and distributed-data tracks, totaling approximately 150 contact hours (e.g., TCH 057, IND 500, TCH 099, GTI 660, GTI 320)
- Authored and refreshed SQL, MongoDB, Kafka, and PostgreSQL exercises with rubrics for IND 500, GTI 660, and related cohorts, matching assessments to learning outcomes
- Unblocked integrator teams on architecture and backend/data; sustained Azure-backed course infrastructure and SCRUM coaching; kept mobile/UX and GTI 320 sessions on track, so students shipped end-to-end features
- Technical Skills:** SQL, MongoDB, Kafka, PostgreSQL, Distributed Databases, Microsoft Azure, Android Studio, UI/UX, SCRUM

IONODES

Laval, QC (Hybrid)

Cloud Developer (Permanent Full-time)

May 2023 – Aug 2023

- Designed a three-tier Auth0 subscription model for organization-scoped IoT access, gating product capabilities per tenant
- Shipped Sentry SDK integrations in C# with Serilog for structured production errors, improving diagnosability for support and engineering
- Deepened ONVIF and WebRTC streaming readiness at IoT scale; orchestrated Azure DevOps with feature flags and repeatable promotions, de-risking releases
- Technical Skills:** C#, .NET, Azure DevOps, Auth0, Sentry, Serilog, UML, Microservices, ONVIF, WebRTC

Intact

Back End Developer (Co-op)

Montréal, QC (Hybrid)

Jan 2022 – May 2022

- Delivered a 6-step signup flow and mobile-app observability on Kotlin/Spring Boot microservices serving 10,000+ active users, accelerating time-to-active coverage
- Implemented Kotlin login and MFA APIs for client mobile apps while preserving policy controls
- Extended the ELK stack with Kibana dashboards and harmonized microservice error codes, shortening incident triage for operations
- **Technical Skills:** Kotlin, Spring Boot, GraphQL, MongoDB, Microservices, Docker Swarm, Jira, Kibana, SAFe

Divine Essence (Union-Nature)

Web Specialist (Contract Part-time)

Dorval, QC (On-site)

Jun 2020 – Dec 2020

- Modernized the bilingual CommerceBuild storefront (FR/EN blog navigation, accessibility, SEO sales labels, custom Add-to-Cart API, 3CX chat-bot), lifting qualified traffic and self-serve support
- **Technical Skills:** HTML, CSS, JavaScript, CommerceBuild, 3CX, SEO

Power Go

Full Stack Web Developer (Co-op)

Montréal, QC (Remote)

Sep 2020 – Apr 2021

- Built a Python/Django/Vue.js/PostgreSQL feed parser unifying XML/JSON/CSV/Excel supplier feeds, integrated Facebook Shop and Marketplace APIs, and shipped a CAD/USD FX-rate service tied to inventory dates, keeping cross-border catalogs in sync without double entry
- **Technical Skills:** Python, Django, Vue.js, PostgreSQL, MongoDB, JavaScript, Git, Jira, Full-Stack Development

Wandrian

Full Stack Developer (Co-op → Part-time)

Montréal, QC (On-site)

May 2019 – Jan 2020

- On the Italiarail stack with Railkey Tech, integrated Braintree and extended purchase APIs while rebuilding the Selenium E2E harness and Django/Pyramid/jQuery internal tools, stabilizing bookings and catching UI regressions early
- **Technical Skills:** Python, Django, Pyramid, JavaScript, jQuery, Selenium, MongoDB, Git, Jira, Test Automation

Algolux

Engineering Intern

Montréal, QC (On-site)

May 2017 – Aug 2017; May 2018 – Aug

2018

- Built C++/CUDA/OpenCV imaging tooling for camera/ISP validation (HDR Nexus 6 rigs, Python RAW-color processing, Flask REST capture) and produced an approximately 12,000-image outdoor vision dataset, shrinking manual validation loops
- **Technical Skills:** Python, C++, CUDA, OpenCV, Flask, Ansible, Linux, Windows

Education

École de technologie supérieure (ÉTS)

Master's Degree, Information Technology Engineering

Montréal, QC, Canada

Aug 2023 – Apr 2026

- Thesis: couples GelSight-style tactile sensing with interactive real-time physics so wear updates friction and shifts texture-space coordinates as the surface geometry changes.

École de technologie supérieure (ÉTS)

Bachelor's Degree, Information Technology Engineering

Montréal, QC, Canada

Sep 2018 – Apr 2023

- AlgoÉTS (ÉTS algorithmic trading club) — President; club infrastructure, web, and trading/data projects.
- **Academic focus:** software engineering, distributed systems, databases, and team-based integrator delivery through ÉTS labs and coursework.